



Start It, Then Repeat

Quick facilitation notes & answer key

Goal

Students see that an EVENT block starts a script and a repeat loop inside it does the repeating. They count the loop, change the event, write an event+loop, and reason about firing the event twice.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to project; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Hook: an alarm (event) starts a routine that repeats — name the GO signal.
2. Project the when-clicked → repeat-3 script; click Bit, count the repeats aloud.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: the event starts it, the loop repeats it.

Answer key (checked by the run-gate)

Ex 1 — 3 times.

Ex 2 — Now you START it by clicking the green flag instead of clicking Bit (the loop runs the same).

Ex 3 — when SPACE key pressed → repeat 4.

Ex 4 (challenge) — One click = 5 moves; two clicks = 10. The event fires twice, so the loop runs twice.

Watch for & differentiate

Common slip: thinking the event itself repeats — it is the loop that repeats, not the event.

Support: point to the event block, say "GO", then trace the loop runs on fingers.

Extend: ask how many moves if you click the flag three times (15).

Success looks like

The student names what starts a script (the event), counts the loop runs (Ex 1), changes the event and describes the difference (Ex 2), writes an event+loop (Ex 3), and reasons that firing the event again re-runs the loop (Ex 4).